STATION:	SALEM 1 & 2
SYSTEM:	Loss of Control Air
TASK:	TCAF a Loss of Control Air
TASK NUMBER:	N1140070401
JPM NUMBER:	15-01 NRC Sim h

ALTERNATE PATH: [X]

K/A NUMBER: APE 065 AA2.06

IMPORTANCE FACTOR: 3.6 (**RO**) 4.2 (**SRO**)

APPLICABILITY:
EO [ ] **RO** [X] **STA** [ ] **SRO** [X]

EVALUATION SETTING/METHOD: Simulator - Perform

REFERENCES:
S2.OP-AB.CA-0001, Rev. 21 Loss of Control Air (both rev checked 8-5-16)
S2.OP-AR.ZZ-0011, Rev. 60, pages 118-128

TOOLS AND EQUIPMENT: None

VALIDATED JPM COMPLETION TIME: 8 minutes

TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS: N/A

Developed By:	**G Gauding** **Instructor**	**Date: 8-5-16**
Validated By:	**M Lutek / W Russell** **SME or Instructor**	**Date: 8-22-16**
Approved By:	**Training Department**	**Date:** 10/11/16
Approved By:	**Operations Department**	**Date:** 10/11/16

ACTUAL JPM COMPLETION TIME:

ACTUAL TIME CRITICAL COMPLETION TIME:

PERFORMED BY:

GRADE: [ ] **SAT** [ ] **UNSAT**

REASON, IF UNSATISFACTORY:

EVALUATOR'S SIGNATURE: **DATE:**

NAME: ____________________

DATE: ____________________

SYSTEM: Loss of Control air

TASK: TCAF Loss of Control Air

TASK NUMBER: N1140070401

SIMULATOR SETUP: **Check 2WG41 is open**

15-01 NRC IC-258

RT-1

MALF: CA0221 #2 STATION AIR CMPRSSR TRIP
REM : CA13D LOCKOUT #1 STATION AIR CMPRSR, Delay 1:30
I/O : AQ01 OVLO COMP 1 TROUBLE ALARM, Fin Val: ON, Delay 1:30

Other setup:
REM: CA15D LOCKOUT #3 STATION AIR CMPRSR
CA11D #1 EMERGENCY AIR COMPRSR IN AUTO Fin Val: MANUAL
CA12M #1 EMRGENCY AIR CMPRSR SWITCH Fin Val: Stop

I/O : AQ01 OVLO COMP 3 TROUBLE ALARM, Fin Val: ON

INITIAL CONDITIONS:

- Unit 2 is operating at 100% power.
- A WG release is in progress from 21 GDT.
- #3 SAC is C/T for scheduled maintenance.
- Unit 1 is operating at 100% power.

INITIATING CUE:

You are the Unit 2 Reactor Operator. Respond to all indications and alarms.

Successful Completion Criteria:
1. All critical steps completed.
2. All sequential steps completed in order.
3. All time-critical steps completed within allotted time.
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made.

Task Standard for Successful Completion:
1. **Isolate Letdown**
2. **Terminate Gaseous Release**
3. **Trip the Rx prior to auto trip on lo lo SG NR level.**

NAME: ______________________

DATE: ______________________

SYSTEM: **Loss of Control Air**

TASK: **TCAF a Loss of Control Air**

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
			Note: The #1 and #3 Compressor Trouble alarms on 2CC1 do NOT have audible or flash capabilities (S2.OP-AR.ZZ-0011 page 120).		
		Simulator Operator: Insert **RT-1** after operator assumes watch.			
			Determines #2 Station Air Compressor (SAC) has tripped, and the Unit 2 Emergency Control Air Compressor (ECAC) has automatically started. **Note**: All 3 SAC supply breakers open is an auto start signal for both ECAC's. The Unit 2 ECAC will auto start, (Unit 1 will not). By design, the ECAC will not load until its control air header pressure lowers to 85 psig. 2A control air header is supplied from #2 ECAC, 2B control air header is supplied from #1 ECAC.		
			Note: The next-to-load SAC (follow) will not immediately start. It requires Station Air header pressure to drop to 5 psig below the follow setpoint (105 psig) for ~ 5 seconds.		
			Refers to the S2.OP-AR.ZZ-0011, Control Console 2CC1, ARP for COMPR 2 TROUBLE.		

NAME: ____________________

DATE: ____________________

SYSTEM: **Loss of Control Air**

TASK: **TCAF a Loss of Control Air**

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	ARP 3.1	IF SAC 2 trips, THEN: A. **COORDINATE** (as necessary) with Unit 1 B. **ENSURE** operating OR **START** at least one SAC IAW SC.OP-SO.SA-0001(Z), Station Air System Operation.	Coordinates with Unit 1 to start at least one SAC IAW SC.OP-SO.SA-0001(Z), Station Air System Operation. **Cue when contacted as Unit 1:** Unit 1 will ensure #1 SAC is placed in service.		
	ARP 3.2	IF AT ANY TIME Station Air pressure cannot be maintained ≥100 psig, THEN **GO TO** S2.OP-AB.CA-0001(Q), Loss of Control Air.	GOES TO S2.OP-AB.CA-0001(Q), Loss of Control Air, when Station Air pressure lowers to 100 psig if AB not entered previously.		
			Simulator Operator: Ensure #1 SAC locks out 1:30 after insertion of **RT-1, then announce twice on plant page: "#1 Station Air Compressor Trip."**		
			Recognizes that no SAC's are running, and enters S2.OP-AB.CA-0001 if not entered previously.		

NAME: ______________
DATE: ______________

SYSTEM: **Loss of Control Air**

TASK: **TCAF a Loss of Control Air**

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
			Note: Approximate event times are listed below **<u>from insertion of RT-1</u>**. 1 min – 1 SAC auto starts. 1 min 30 sec – 1 SAC trips. 3 min 50 sec – 2 ECAC begins loading. 4 min 10 sec - 2B Control Air header pressure reaches 80 PSIG, and SA low pressure alarm. 4 min 25 sec – CA low pressure alarm. 6.5 min – 21-24BF19 Feed Reg Valves begin closing. 7 min 10 sec – G-15 OHA ADFCS TRBL.		
	AB.CA 3.1	**INITIATE** Attachment 1, Continuous Action Summary.	Initiates Attachment 1, Continuous Action Summary. **Cue <u>IF</u> Required:** If CAS Step 8.0 action is being taken to refer to Attachment 2 for valve fail positions and for further action to be taken, then: **Cue: The CRS will refer to Attachment 2**		
	3.2	EITHER of the following conditions exist? • ALL Station Air Compressors stopped • EITHER Station Air Header <100 psig	Determines all SAC's are stopped or determines Station Air Header pressure is <100 psig.		

NAME: ____________________

DATE: ____________________

SYSTEM: **Loss of Control Air**

TASK: **TCAF a Loss of Control Air**

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	3.3	IF a loss of **1**F Group Bus has occurred, AND **1**H Group Bus is available, THEN **SEND** an Operator to restore power to #1 SAC Auxiliary Oil pump by transferring **11** Turbine West 230V Control Center IAW Attachment 14, Section 1.0.	Determines a loss of 1F Group Bus has not occurred.		
	3.4	IF a loss of **2**F Group Bus has occurred, AND **2**H Group Bus is available, THEN **SEND** an Operator to restore power to #2 SAC Auxiliary Oil pump by transferring 21 Turbine West 230V Control Center IAW Attachment 14, Section 2.0.	Determines a loss of 2F Group Bus has not occurred.		
	3.5	IF a loss of **1**H Group Bus has occurred, AND **1**F Group Bus is available, THEN **SEND** an Operator to restore power to #3 SAC Auxiliary Oil pump by transferring 11 Turbine East 230V Control Center IAW Attachment 14, Section 3.0.	Determines a loss of 1H Group Bus has not occurred.		

NAME: ______________________

DATE: ______________________

SYSTEM: **Loss of Control Air**

TASK: **TCAF a Loss of Control Air**

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	3.6	**START** or have Unit One Control Room **START** Next to Load Station Air Compressor.	Determines no SAC's are available. **Cue if required:** If Unit 1 is contacted, report that #3 Station Air Compressor is tagged out and #1 Station Air Compressor has tripped, and that an operator has been dispatched to the Station Air Compressors.		
	3.7	Either Station Air Header indicating less than100 psig?	Determines both Station Air Headers are indicating less than100 psig.		
	3.8	Attempt to **START**, or have Unit One Control Room **START**, remaining Station Air Compressor.	Determines no SAC's can be started.		
	3.9	**NOTIFY** the CRS/SM to evaluate use of temporary air compressors.	**Cue:** CRS will evaluate use of temporary air compressors.		
	3.10/3.11	2A Control Air Header less than or equal to 88 psig?	Determines 2A control air header pressure is: less than or equal to 88 psig and determines #2 ECAC is in service OR >88 psig.		

NAME: ____________________

DATE: ____________________

SYSTEM: **Loss of Control Air**

TASK: **TCAF a Loss of Control Air**

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	3.12/3.13	2B Control Air Header less than or equal to 88 psig?	Determines 2B control air header pressure is: less than or equal to 88 psig and contacts Unit 1 to place #1 ECAC in service OR >88 psig. **Cue:** When Unit 1 is contacted to start #1 ECAC, state: **" # 1 ECAC has tripped. "**		
	3.14	**SEND** operators to: • **LOCATE** AND **ISOLATE** ANY air system leaks. • **INVESTIGATE** cause of ANY air compressor trip/trouble.	Dispatches operators to: • **LOCATE** AND **ISOLATE** any air system leaks. • **INVESTIGATE** cause of air compressor trip/trouble.		
	3.15	IF Station Air is lost, AND EITHER of the following are in service: • Spent Fuel Pool Transfer Pool Weir Gate Seals • SG Nozzle Dams THEN: A. **EVACUATE** personnel from areas subject to leakage/flooding. B. **INITIATE** monitoring of Fuel Pool or Reactor Vessel levels for the need to makeup due to leakage.	Determines Station Air is lost, and neither of the following are in service: • Spent Fuel Pool Transfer Pool Weir Gate Seals • SG Nozzle Dams		

NAME: ____________
DATE: ____________

SYSTEM: **Loss of Control Air**
TASK: **TCAF a Loss of Control Air**

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	3.16	IF cooling water to the Station or Emergency Air Compressors is lost, THEN **INITIATE** the applicable portion of Attachment 6.	Determines cooling water to the Station or Emergency Air Compressors is not lost.		
	3.17	BOTH 2A and 2B Control Air Headers less than 80 psig?	Determines both 2A and 2B Control Air Headers are not <80 psig.		
	3.18	ANY Excess Flow Check Valve closed as indicated by at least one of the following? (Attachment 5, Excess Flow Check Valve Table) **[INPO IER L2 15-16]** • Groups of valves repositioning with normal air pressure indicated in Control Room • Local observation of low pressure downstream of an Excess Flow Check Valve	Determines NO Excess Flow Check Valve closed.		
	3.19	2B Control Air Header less than 80 psig?	Determines 2B Control Air Header is <80 psig. Alternate Path actions start here.		
	3.57	**CONTINUE**			
*	3.58	**CLOSE** the following valves: A. 2CV3, LTDWN ORIFICE ISOL VALVE B. 2CV4, LTDWN ORIFICE ISOL VALVE C. 2CV5, LTDWN ORIFICE ISOL VALVE D. 2CV2 LETDOWN LINE ISOL VALVE E. 2CV277 LTDWN LINE ISOL VALVE F. 2CV7 LTDWN HX INLET VALVE	Closes the following valves: A. 2CV3, LTDWN ORIFICE ISOL VALVE B. 2CV4, LTDWN ORIFICE ISOL VALVE C. 2CV5, LTDWN ORIFICE ISOL VALVE D. 2CV2 LETDOWN LINE ISOL VALVE E. 2CV277 LTDWN LINE ISOL VALVE F. 2CV7 LTDWN HX INLET VALVE		

NAME: ____________________
DATE: ____________________

SYSTEM: **Loss of Control Air**
TASK: **TCAF a Loss of Control Air**

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
*	3.57	**VERIFY** ANY Liquid or Gaseous release is stopped by ensuring the following valves CLOSED: • 2WL51, TO CIRC WTR DISCHARGE • 2WG41, GAS DECAY TK TO PLANT VENT	Determines 2WG41, GAS DECAY TK TO PLANT VENT is open, and shuts 2WG41, GAS DECAY TK TO PLANT VENT. Verifies 2WL51 is shut.		
*	CAS 6.0	IF **AT ANY TIME** Station Air is lost, THEN: • **INITIATE** monitoring of 21-24BF19 SG FW CONT VALVE, operation. • IF ANY BF19 SG FW CONT VALVE closes AND Applicable SG water level CANNOT be maintained, THEN: A. **TRIP** the Reactor B. **CONTINUE** with this procedure C. **GO TO** 2-EOP-TRIP-1, Reactor Trip Or Safety Injection.	Identifies all BF19 valves closing. Trips the Reactor.		

NAME: ____________________
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SYSTEM: **Loss of Control Air**
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*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	TRIP-1		Performs Immediate Actions of EOP-TRIP-1: Confirms the Rx trip Trips the Main turbine Checks at least 1 4KV vital bus energized Determines SI is not initiated or required. **Cue after Immediate Actions have been performed: The CRS and another operator will continue performing the EOPs. You are to continue in current procedure.**		
			Terminate JPM once Rx has been tripped, letdown isolated, and the WG release terminated.		

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

✓ WA 1. Task description and number, JPM description and number are identified.

✓ WA 2. Knowledge and Abilities (K/A) references are included.

✓ WA 3. Performance location specified. (in-plant, control room, or simulator)

✓ WA 4. Initial setup conditions are identified.

✓ WA 5. Initiating and terminating Cues are properly identified.

✓ WA 6. Task standards identified and verified by SME review.

WA 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).

WA 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. 21 Date 8/22/16

WA 9. Pilot test the JPM:
   a. verify Cues both verbal and visual are free of conflict, and
   b. ensure performance time is accurate.

________ 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.

________ 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: MLVTEK Date: 8/22/16

SME/Instructor: W. Russell Russell Date: 8/22/16

SME/Instructor: ________________ Date: ________

INITIAL CONDITIONS:

- Unit 2 is operating at 100% power.
- A WG release is in progress from 21 GDT.
- #3 SAC is C/T for scheduled maintenance.
- Unit 1 is operating at 100% power.

INITIATING CUE:

You are the Unit 2 Reactor Operator. Respond to all indications and alarms.